An uncooked vegan diet shifts the profile of human fecal microflora: computerized analysis of direct stool sample gas-liquid chromatography profiles of bacterial cellular fatty acids.

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The effect of an uncooked extreme vegan diet on fecal microflora was studied by direct stool sample gas-liquid chromatography (GLC) of bacterial cellular fatty acids and by quantitative bacterial culture by using classical microbiological techniques of isolation, identification, and enumeration of different bacterial species. Eighteen volunteers were divided randomly into two groups. The test group received an uncooked vegan diet for 1 month and a conventional diet of mixed Western type for the other month of the study. The control group consumed a conventional diet throughout the study period. Stool samples were collected. Bacterial cellular fatty acids were extracted directly from the stool samples and measured by GLC. Computerized analysis of the resulting fatty acid profiles was performed. Such a profile represents all bacterial cellular fatty acids in a sample and thus reflects its microflora and can be used to detect changes, differences, or similarities of bacterial flora between individual samples or sample groups. GLC profiles changed significantly in the test group after the induction and discontinuation of the vegan diet but not in the control group at any time, whereas quantitative bacterial culture did not detect any significant change in fecal bacteriology in either of the groups. The results suggest that an uncooked extreme vegan diet alters the fecal bacterial flora significantly when it is measured by direct stool sample GLC of bacterial fatty acids.